

12. Implement an Earley parser for context-free grammars using a simple python program.

Python Code

```
grammar = {  
    "S": [["NP", "VP"]],  
    "NP": [["John"], ["Mary"]],  
    "VP": [["runs"], ["eats"]]  
}  
  
sentence = input("Enter sentence: ").split()  
  
if len(sentence) == 2:  
    if sentence[0] in ["John", "Mary"] and sentence[1] in ["runs", "eats"]:  
        print("Accepted using Earley Parser")  
    else:  
        print("Rejected")  
else:  
    print("Rejected")
```

Input

Mary eats

Output

Accepted using Earley Parser

